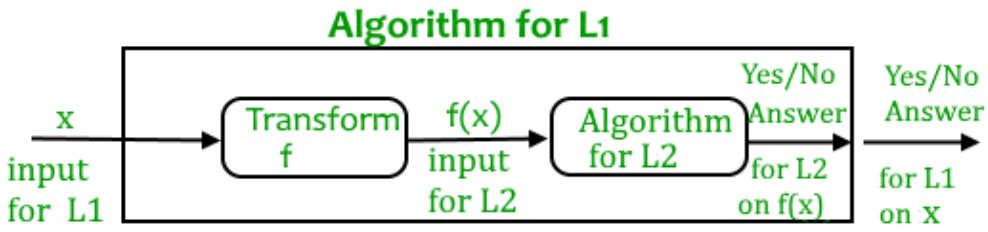
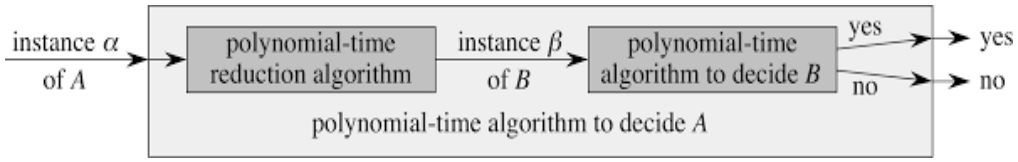
Polynomial Time Reduction

- In computational complexity theory, a polynomial-time reduction is a method for solving one problem using another
- If both the time required to transform the first problem to the second, and the number of times the subroutine is called is polynomial, then the first problem is polynomial-time reducible to the second.
- We have two problems, A and B, and we know problem B is a P class problem.
- If problem A can be reduced, or converted to problem B, and this reduction takes a polynomial amount of time, then we can say that A is also a P class problem (A is reducible to B).
- We say that Decision Problem L1 is Polynomial time Reducible to decision Problem L2 ($L1 \leq_p L2$) if there is a polynomial time computation function f such that of all x, $x \epsilon L1$ if and only if $x \epsilon L2$.



- We can show that decision problem A is in P if there is a poly-time-alg that reduces A to a decision problem B which is known to be in P.
- That is, each instance $\alpha \in A$ is converted to an instance $\beta \in B$ such that B's algorithm delivers the answer to α.


instance α
of A
polynomial-time
reduction algorithm
instance β
of B
polynomial-time
algorithm to decide B
yes
no
yes
no
polynomial-time algorithm to decide A